This report reveals a stark divide in European '*Education Resilience.*' While tertiary education acts as a *20+ point* shield in markets like *Slovakia*, structural inversions in countries like *Turkey* show that graduates can sometimes face higher risks than primary-level workers, particularly among youth, where the gap reaches *-13.40 points.*

Geographic Distribution

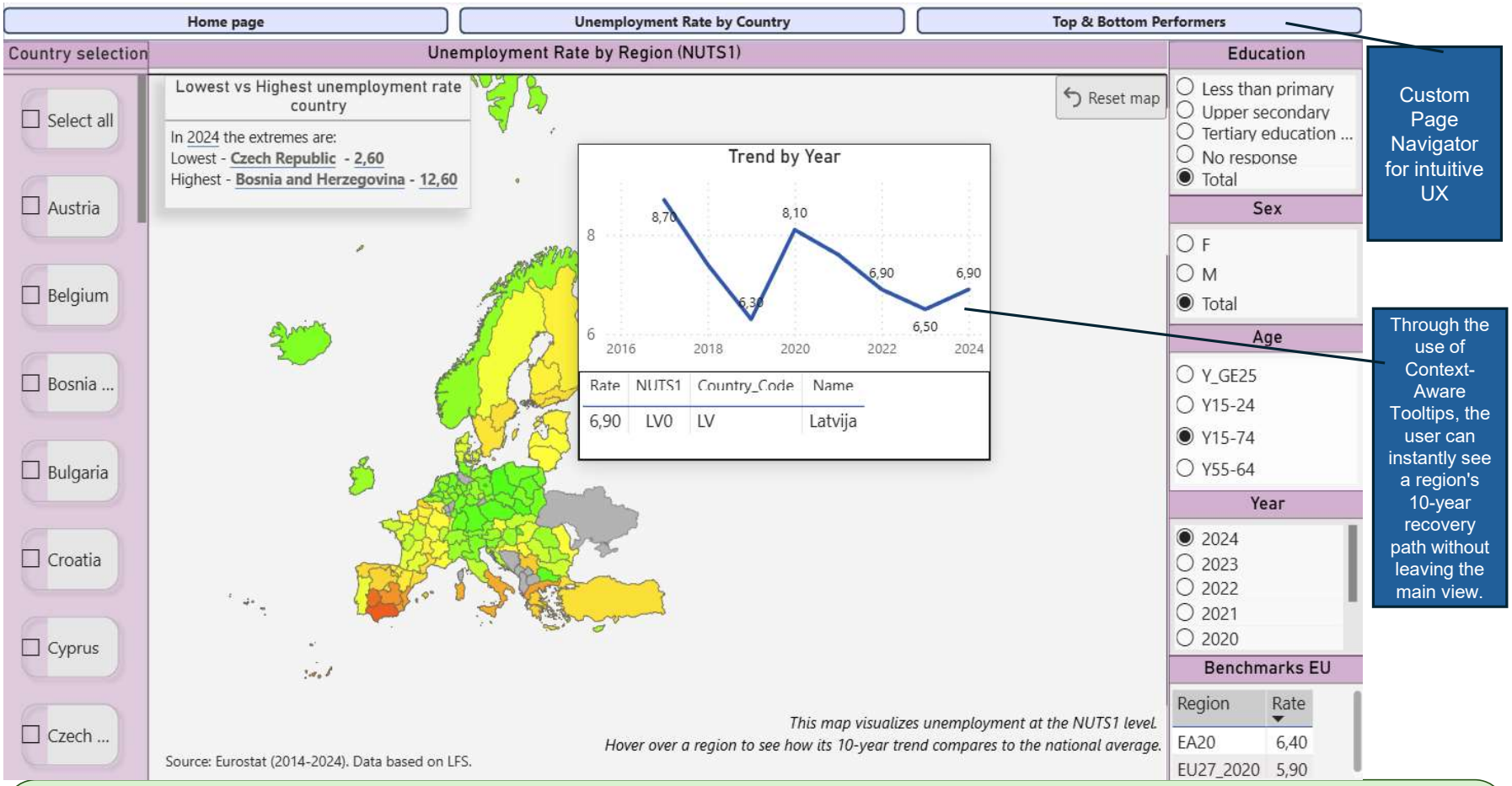
Demographic Deep-Dive

The Education Premium

Technical Appendix

EU27 for 2024		Structural Inversion Detected	Youth Vulnerability Gap
5,90 EU27	▼ -3.3% Change YoY%	-13,40 Turkey Education..	9,90 Rate Gap EU
12,60 Max	2,60 Min		

The pages were grouped into 3 theme pages, home page and technical appendix page.
The 3 categories accessible through the home pages have navigation bars of all the pages that are in the category. In total, there are 7 visual insight pages split into the 3 categories.



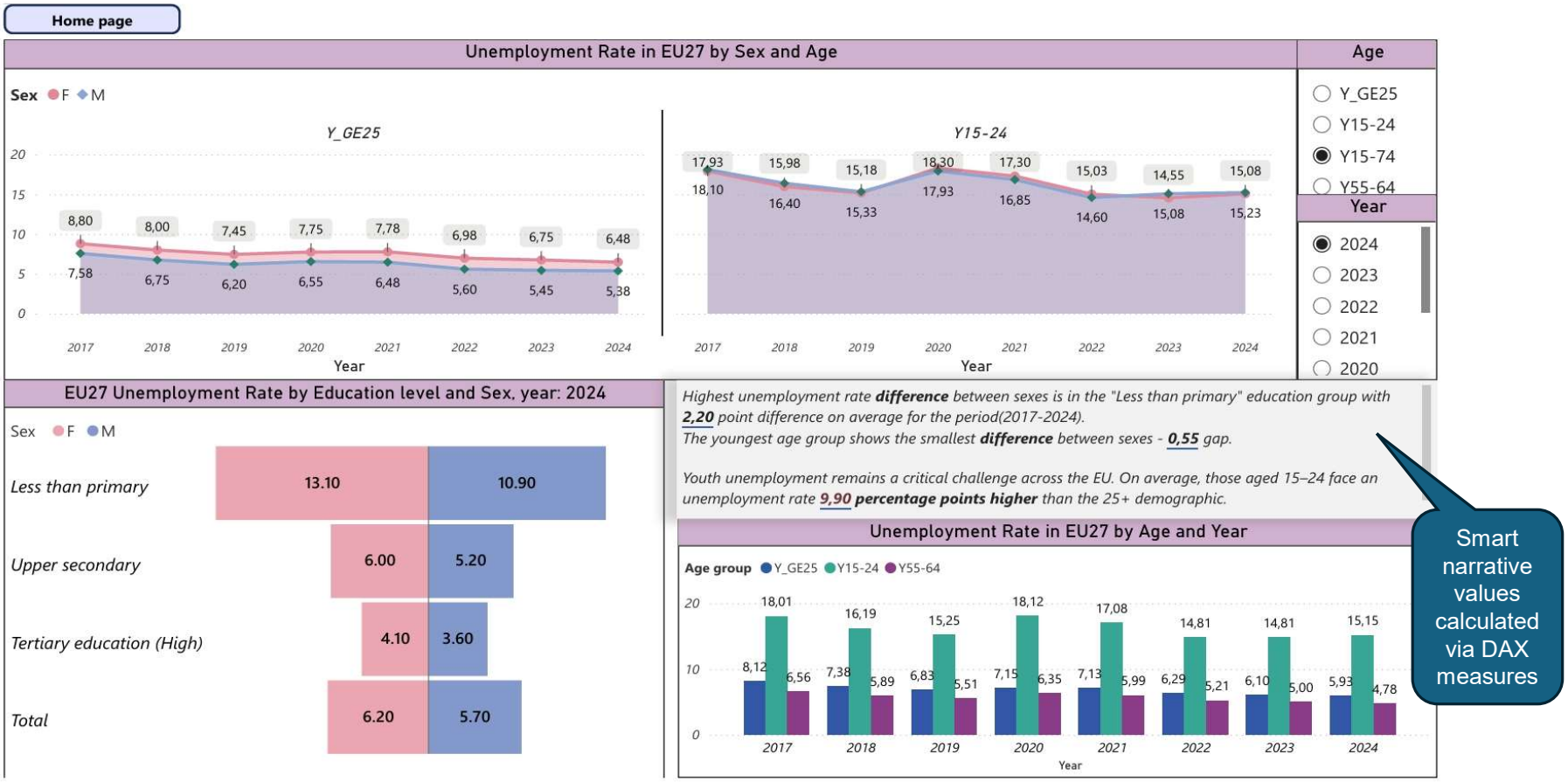
Geographic Disparity: Southern Europe faces 3x higher youth unemployment than the North.
Through the use of Context-Aware Tooltips, the user can instantly see a region's 10-year recovery path without leaving the main view.
Color code is dynamic depending on unemployment rate, red/orange areas have highest unemployment.
Dynamic narrative on the top left shows extremes. It uses smart values and DAX to display current data for the selected slicers.
EU map was created as custom map with data in GeoJSON file format available at Eurostat geodata.

The first chart shows the distribution of countries around the benchmark (EU27 average).
EU Performance: Steady *7-year decline*, interrupted only by the *2020 COVID-19 pandemic.*
Significant outliers:

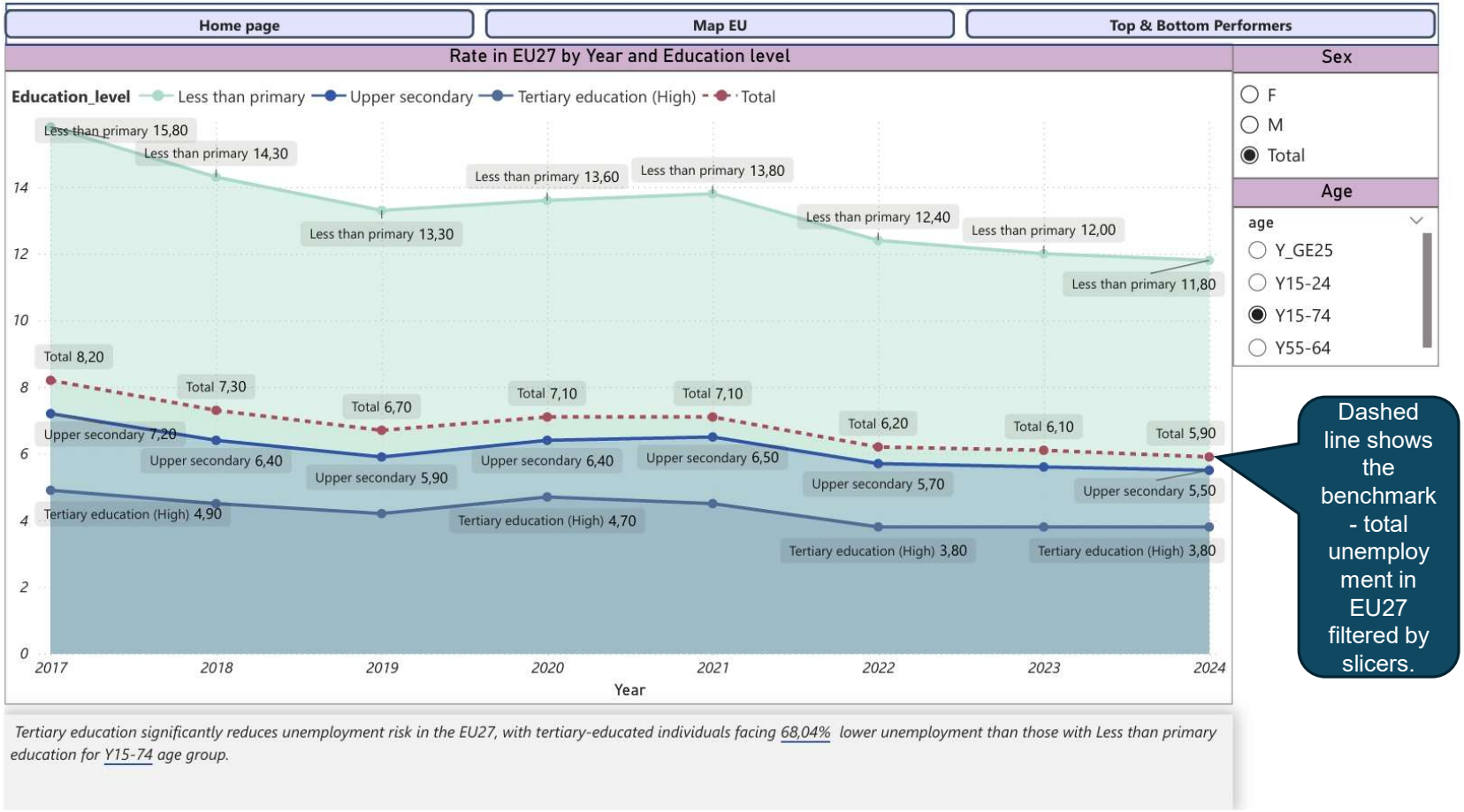
- *Lowest : Czech Republic-* ***2.60***
- *Highest (EU) : Spain* ***-11.40*** ; *highest overall:* ***Bosnia*** (*candidate country*) ***–12.60***

The bottom chart shows year-over-year change in percentage(YoY%), which shows change in rate.

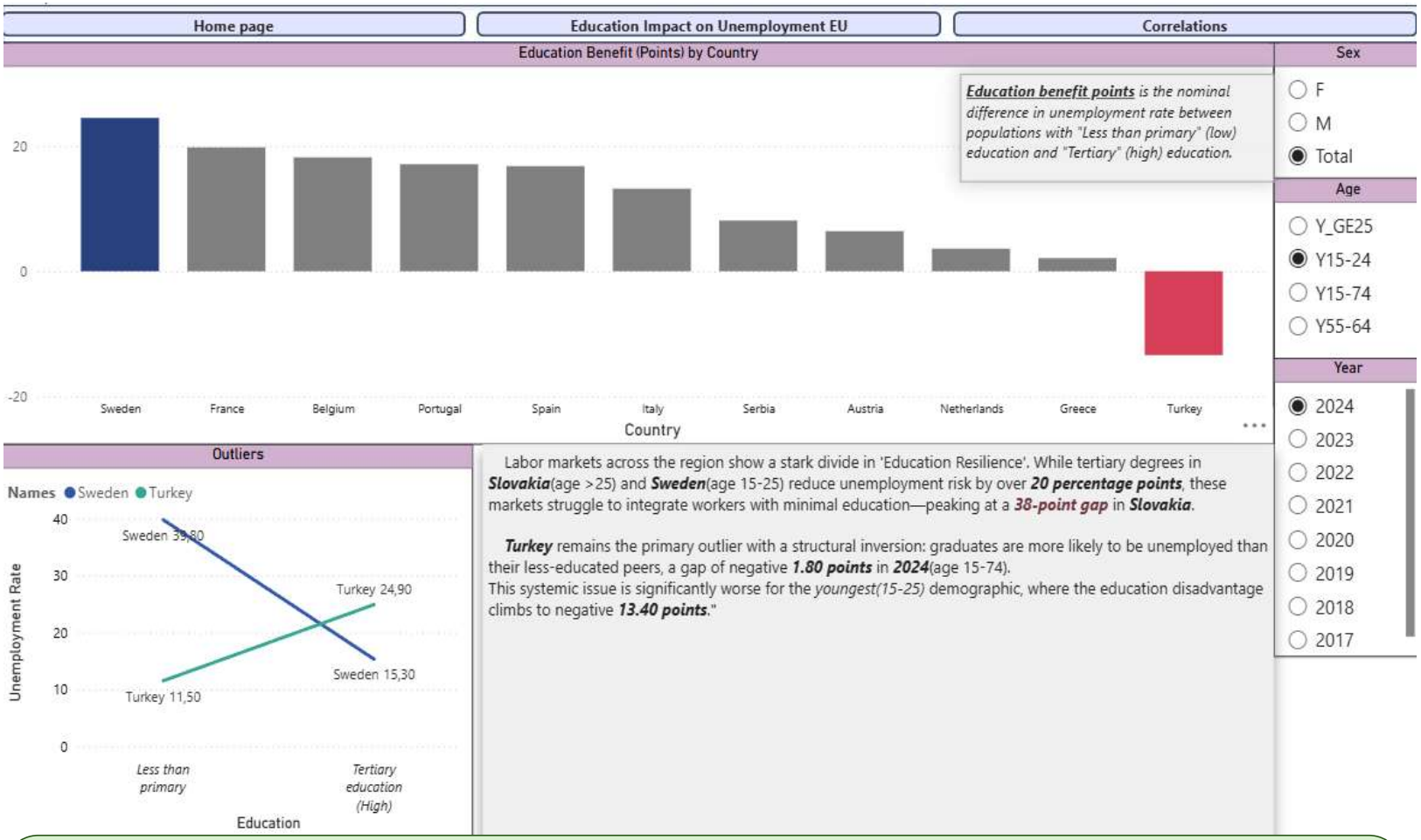
Top 5 vs bottom 5 countries show the stark *difference in unemployment across EU*.
The region chart based on *NUTS 2* reveals that some regions can have very high unemployment despite the country having a reasonable performance as a whole.
Three regions in ***France*** have high unemployment*:* ***FR4, FR3, FR1 ~ 17.00,*** *contrasting the country as a whole* ***- 7.5.***



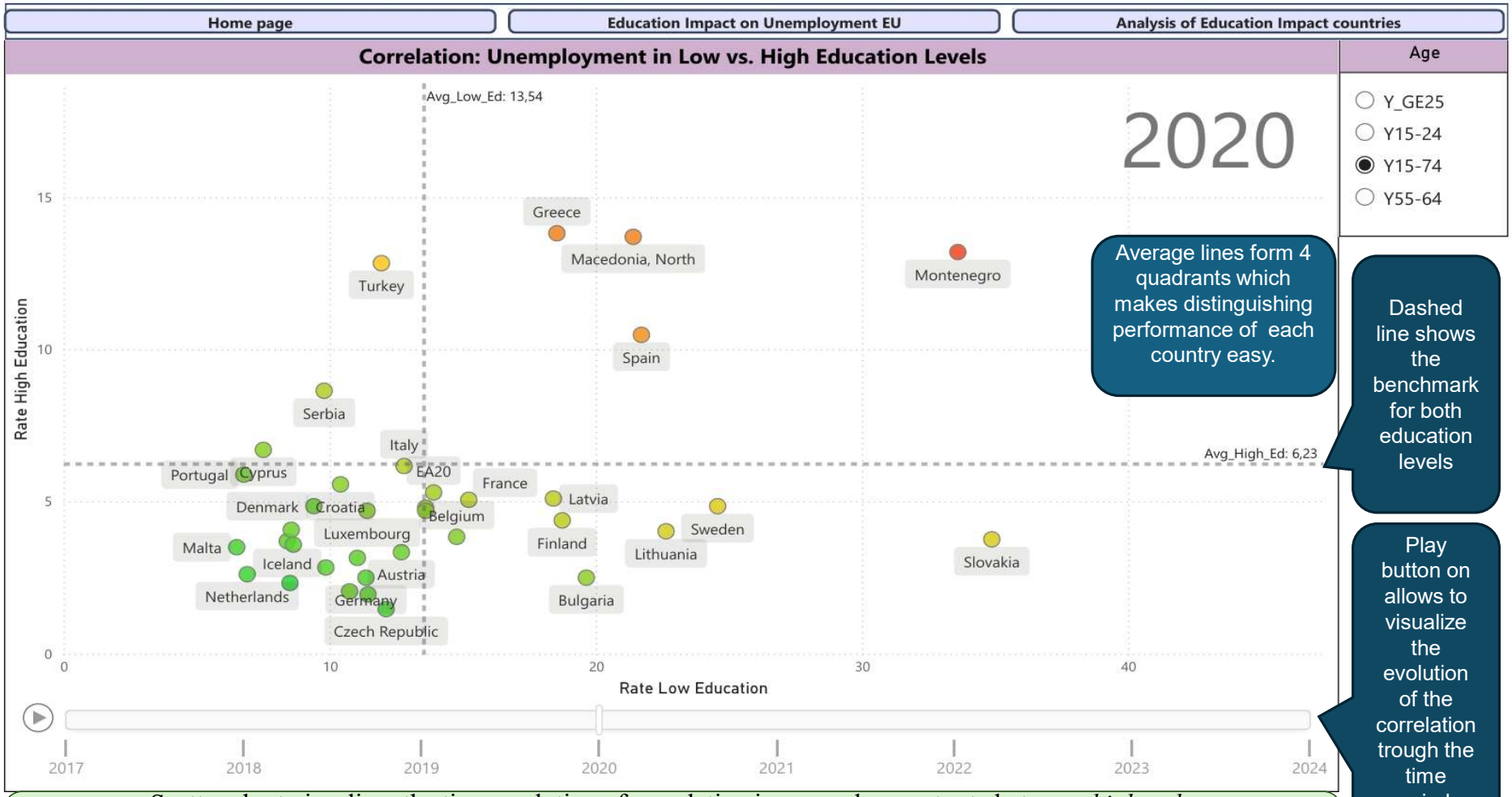
The analysis shows the unemployment difference between sexes and ages. Sex disparity is mostly observed among low education group – 2,2 points.
The *most significant difference* is observed among *age groups*.
Youth unemployment is the biggest challenge for EU.



Education level has a significant effect on unemployment levels.
Higher education(Tertiary education) is the *best protection* against unemployment in the EU as a whole. Higher education people are *almost 3 times* less likely to be unemployed.



Education benefit points is a measure that is calculated as the nominal difference in unemployment rate between education groups. The visual displays the starkest *disparity between education groups* and the structural trend that most often favours Higher education.
The *Slope chart* in the bottom-left visualizes the *extremes* filtered using DAX measure as a filter.



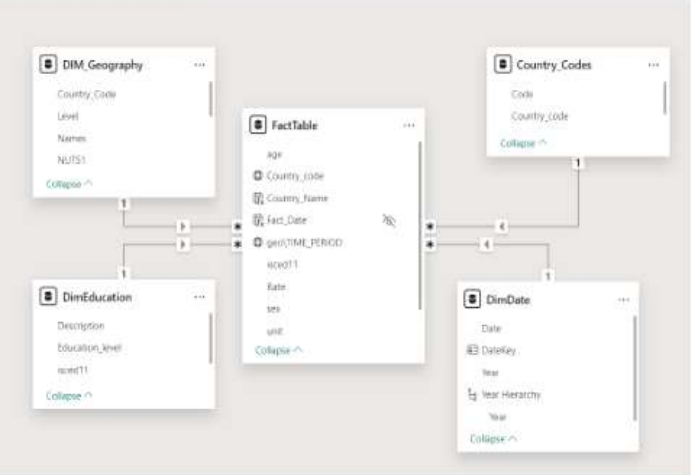
Average lines form 4 quadrants which makes distinguishing performance of each country easy.

Dashed line shows the benchmark for both education levels

Play button on allows to visualize the evolution of the correlation trough the time periods

Scatter chart visualizes the time evolution of correlation in unemployment rate between *high vs low education levels* for each country and each year.
Bottom-left quadrant shows the best performing countries in the time frame.
Top-right quadrant holds the worst performing countries.

Home page



The report utilizes a Star Schema with a central Fact Table (Unemployment Data) linked to optimized Dimension tables (Geography, Education, Date, Country codes). This structure ensures high performance and accurate cross-filtering.

Data Source: Eurostat Labour Force Survey (LFS). [lfst_r_lfu3rt]

Data Architecture

The Eurostat *EU labour force survey (LFS) data* had to be cleaned and transformed into long data structure suitable for Star Schema.

- Data for *2025* is still not published as of March 2026.
- *Unpivoting columns* year and education level.
- *Non-country **aggregate values** (like EA20)* had to be kept into the dimensions table, since the nature of the data do not allow the calculation of aggregates without *weighted values* (unemployment rate is function of population). The aggregate values were left in the Fact Table for the same reason. A column was created to distinguish types of aggregate data by geographical level (Area, country, region). The reason for that is to prevent *double-counting*.

UI/UX

- ***EU map*** was created as custom map with data in *geoJSON* file format available at Eurostat geodata.
- ***Navigation***: Page Navigator is used for easy access guided experience. Pages are grouped into topics to keep the story and insight for the viewer on focus.

Hidden pages are used to create a user experience via Page Navigator.

- ***Smart Narratives:*** Created to provide automated, data-driven commentary. Some are using DAX measures or Values to display dynamic output based on selected slicers.
- ***Conditional Formatting*** is used on some of the charts to make readability and better display the insight.

DAX Showcase

One notable *measure* was needed for the *'Education Shield'* calculation. Education Benefit Points show the nominal difference in unemployment rate between low and high education groups.

DAX measure was created that accounts for null values across levels. It also had to make sure the value is calculated on country level, excluding the aggregate values in the data(like EA20).

This ensures that only countries with complete data for both primary and tertiary levels are compared, preventing skewed outlier results.

DAX measure:

```
Education Benefit (Points) =
VAR LowRate =
    CALCULATE(
        [AVG Unemployment Rate],
        'DimEducation'[Education_level] = "Less than primary",
        DIM_Geography[Level]="Country"
    )
VAR HighRate =
    CALCULATE(
        [AVG Unemployment Rate],
        'DimEducation'[Education_level] = "Tertiary education (High)",
        DIM_Geography[Level]="Country"
    )

RETURN
-- eliminatoing nulls
IF(
    ISBLANK(LowRate) || ISBLANK(HighRate),
```

Technical appendix page shows some of the work steps and some of the DAX logic behind the project.